
Context Fatigue Is In-Context Comfort: Multi-Turn Confidence Signatures Without Performance Loss

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Abstract

As a multi-turn conversation accumulates, instruction-tuned LLMs show two vivid “context-fatigue” signatures: output entropy collapses several-fold, and the last token’s attention drifts off the system prompt and the current query onto recent context. We reproduce both across Qwen-2.5-7B, Llama-3.1-8B, Gemma-2-9B, and the full OLMo-2-7B post-training chain—and then show, with a design that other context-rot studies cannot use, that they are *not* a competence decay. Our key methodological choice is to accumulate **individually localized tasks** (self-contained multiple-choice cases) rather than one long task, so the information needed for the current answer is always at the current query and never diluted across length. This *structurally rules out attention dilution / lost-in-the-middle* as the cause. With dilution removed, the signatures persist but performance does not move: per-case accuracy is flat-to-rising with context fill ($r \approx 0$ even when in-context learning is deterred with a random-subject stream), checkable instruction adherence stays at ceiling, and—conditioning on fill—the cases the model gets *wrong* are the ones where it *fixates* on the current query, while correct answers spread attention onto the accumulated context (the per-case fingerprint of in-context learning). The entropy collapse vanishes on organic WildChat dialogue (0.99 within-conversation late/early ratio) and tracks conversation *homogeneity*, not length. An OLMo-2 dose-response shows the collapse is a baseline-confidence *level shift* installed mostly by supervised fine-tuning, and a causal test of the boundary-detector feature blamed for it does not rescue the model. We argue “context fatigue” as measured is in-context comfort, not decay; the genuine hazard is a widening confidently-wrong gap: confidence climbs $0.85 \rightarrow 0.95$ with fill ($r = +0.72$), on wrong answers as much as right, at flat accuracy.

1 The claim under test

A recurring worry about long conversations is *context rot*: as turns accumulate, the model is said to degrade. The phenomenology is real and easy to elicit. On a repeating multiple-choice task, instruction-tuned models become dramatically more confident as context fills—next-token entropy drops 3–4 $\times$ on Qwen-2.5-7B and Llama-3.1-8B, in both question orders—and the last token’s attention shifts systematically: away from the system prompt and the current query, toward the most recent prior cases, with the attention distribution diffusing. The tempting reading is that the model is being worn down: it stops reading its instructions, fixates on recent text, and its collapsing entropy is overconfidence that should translate into errors and into sycophantic capitulation when challenged.

This abstract tests that reading and finds it inverted. The signatures are robust and replicate across four model families, but they are signatures of *in-context learning*, not of degradation. We make the case with a design that separates accumulation from the confound that dominates the existing literature.

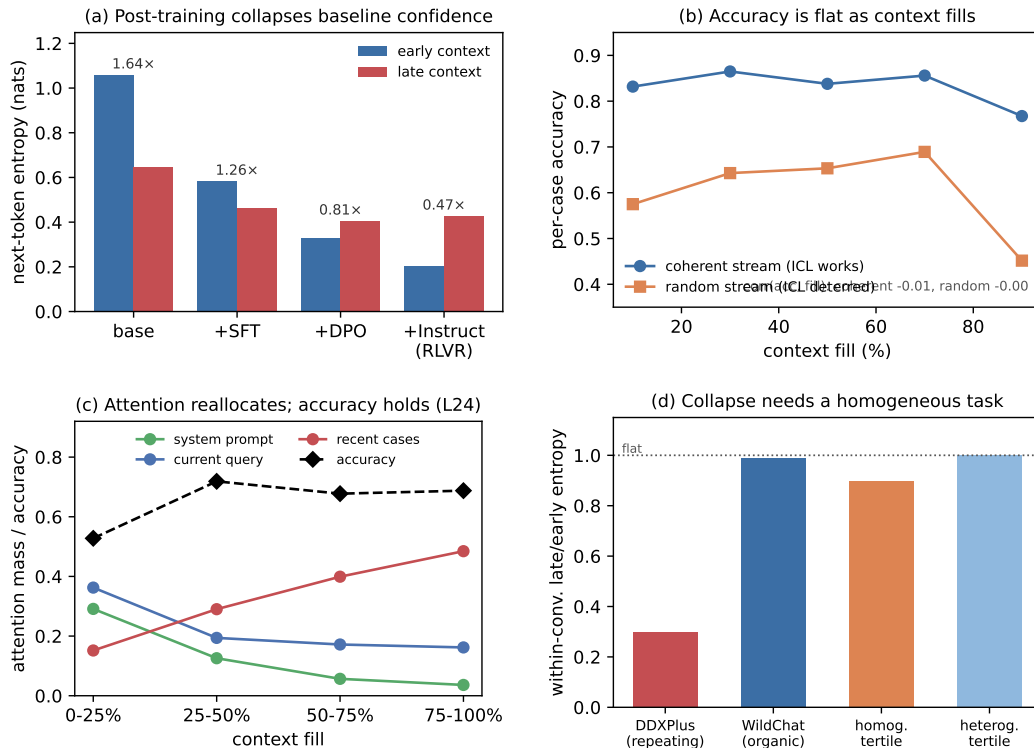


Figure 1: **The signatures are real but performance is not affected.** (a) Across OLMo-2’s post-training chain, early-context (baseline) entropy collapses monotonically ($\sim 5\times$); the within-context early/late ratio (annotated) falls from collapse ($1.64\times$, base) to mild reversal ($0.47\times$, Instruct). (b) Per-case accuracy is flat as context fills, even when in-context learning is deterred by drawing each question from a random MMLU subject ($r\approx 0$). (c) At layer 24 (OLMo-2-Instruct), attention mass moves off the system prompt and onto recent cases monotonically with fill, while per-case accuracy holds. (d) The within-conversation entropy collapse appears only with a homogeneous repeating task (DDXPlus), not on organic WildChat dialogue, and tracks homogeneity rather than length.

2 Design: localized tasks rule out attention dilution

Most context-rot evaluations embed the answer-bearing information inside *one long task*—a long document, a buried “needle,” a growing codebase. There, longer context mechanically *dilutes* the relevant tokens among irrelevant ones, so any degradation conflates two things: the cost of accumulation per se and the cost of the target being harder to attend to (“lost in the middle,” Liu et al., 2023). These cannot be told apart in that setup.

Our harness instead accumulates **individually localized tasks**: a stream of self-contained 5-option medical cases (DDXPlus, Tchango et al., 2022) or discrete factual questions (MMLU, Hendrycks et al., 2021). Each answer depends only on the *current* question, which always sits at the end of the context; prior turns are never required to answer it. Accumulation therefore grows the context without ever diluting the information the current answer needs. **This structurally removes attention dilution as a candidate mechanism**—if performance is flat here, it cannot be rescued or explained by dilution—and lets us read accumulation effects cleanly. A reversed-question-order control additionally removes any difficulty-ordering confound (the entropy collapse is order invariant). Because a 4–32k window holds only tens of cases, we pool many independent accumulation sessions (different case and option orders) to power per-case analysis.

3 The signatures are real—and they replicate

The attention reallocation is not an artifact of our attention reconstruction: we replicate OLMo-2’s post-RoPE last-token attention and validate it against the model’s own `output_attentions` to $\max |\Delta| \approx 1.4e-3$. With context fill it produces strong, monotone, per-case correlations at every

Table 1: Per-case, conditioning on fill (OLMo-2-Instruct, layer 24): wrong cases *fixate* on the current query. Current-query attention mass by correctness, heads averaged within case (no head-level pseudo-replication); seeded bootstrap 95% CIs.

fill bin	0–25% ($n=36$)	25–50% ($n=32$)	50–75% ($n=31$)	75–100% ($n=16$)
correct	0.349	0.193	0.168	0.153
wrong	0.377	0.197	0.179	0.182
Δ (wrong–correct)	+0.028	+0.005	+0.010	+0.029
95% CI	[−.08, +.13]	[−.01, +.02]	[−.00, +.02]	[+.01, +.05]

probed layer (layer 24: system $\leftrightarrow$ fill -0.93 , recent $\leftrightarrow$ fill $+0.89$, current $\leftrightarrow$ fill -0.71 , entropy $\leftrightarrow$ fill $+0.95$; Fig. 1c). The DPO and Instruct checkpoints are indistinguishable, so the reallocation is present from the SFT/DPO stage, not introduced by the final RLVR step. The entropy collapse likewise replicates across model families. By every visual diagnostic, this is a real, strongly-measurable effect that grows with context—exactly the thing one would report as “context rot.”

4 ...but there is no performance cost

Accuracy is flat, and stays flat when ICL is removed. Per-case accuracy does not fall as context fills; it is worst at the cold start and warms up ($\text{corr}(\text{correct}, \text{fill}) = +0.11$ Instruct, $+0.07$ DPO). The obvious objection is that a homogeneous repeating task lets in-context learning mask a hidden decay. We remove that crutch: in a *random-subject* stream each question is drawn uniformly from all 56 MMLU subjects, so prior context carries no predictive information about the next question—only the answer *format* is learnable. Accuracy is still flat across the first $\sim 80\%$ of context ($0.57 \rightarrow 0.69$, $\text{corr} = -0.00$), versus a coherent single-subject stream (0.84 , $\text{corr} = -0.01$); the lower level just reflects harder questions, not a slope (Fig. 1b). These nulls are *bounded*: the 95% interval on upper-minus-lower-half accuracy excludes declines beyond 7.3 points (coherent, $n=483$) and 9.2 (random, $n=344$). The $\sim 80\%$ restriction reflects a real top-bin dip on the random stream (-0.19 [$-0.37, -0.00$], $n=31$). With dilution already excluded by design, neither accumulation nor the loss of ICL produces decay.

Instruction adherence holds. Looking for a competence-preserved, compliance-degraded dissociation, we add a checkable “canary” system instruction (begin every reply with a marker; end with a tag) and track obedience versus fill across baseline, forced-history, and refresh arms. Adherence is perfect throughout ($\text{corr}(\text{violation}, \text{fill}) = 0$) with accuracy flat—an honest null: the canaries were easy enough to sit on a ceiling, so this bounds rather than proves the effect, but it shows no adherence decay where the degradation reading predicts one.

The per-case mechanism inverts the story. Conditioning on context fill, the cases OLMo-2 answers *wrong* have *higher* current-query attention and lower, more peaked attention entropy than the ones it gets right (Table 1). The effect is small (pooled $\Delta = +0.045$ [$-0.003, +0.097$], $n=115$, significant only in the fullest bin), but its sign is positive in every fill bin and at every probed layer (8/16/24/31): nowhere do wrong answers *under-attend* the current query, as the degradation reading predicts. Errors are associated with *fixating* on the query; correct answers spread attention onto the accumulated cases—i.e. onto the in-context examples. The “rot” (system erosion, recency bias, diffusion) and the within-context predictor of a correct answer point in *opposite* directions. More contextual attention is the fingerprint of the ICL that keeps accuracy up, not a symptom of decay.

5 The entropy collapse is in-context comfort, not depth

If the collapse were a generic depth effect it should survive on heterogeneous dialogue. It does not. On 150 organic WildChat conversations (Zhao et al., 2024) (Qwen-2.5-7B, teacher-forced over the real history), within-conversation entropy is flat with depth: median late/early ratio 0.99, within-conversation $\text{corr}(\text{entropy}, \text{depth})$ median -0.02 , and 52% of conversations trend down—a coin flip (Fig. 1d). The 3–4 $\times$ DDXPlus collapse simply does not appear once the task stops repeating. Partitioning WildChat by its *own* inter-turn homogeneity (format and dataset held constant) isolates the driver: the entropy slope correlates with homogeneity (partial $r = -0.151$ controlling for length), homogeneous conversations collapse (0.897 late/early) while heterogeneous ones are flat

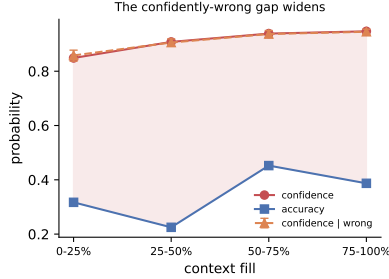


Figure 2: **The confidently-wrong gap widens with fill** (pooled DDXPlus MCQ, $n=154$). Shaded area: calibration gap; error bars: bootstrap 95% CIs.

(1.001), and because homogeneous conversations are *longer*, length works against the effect. The collapse is the model growing comfortable with a repeating task—in-context learning sharpening a distribution—not fatigue with depth.

6 It is an instruction-tuning level shift, and the proposed mechanism does not hold

Replaying one fatigue probe across OLMo-2-7B’s base→SFT→DP0 →Instruct chain (identical cases and orders, so differences are attributable only to post-training) turns a single correlational base-vs-IT pair into a graded curve. Early-context entropy collapses monotonically, $1.06 \rightarrow 0.58 \rightarrow 0.33 \rightarrow 0.20$ ($\sim 5\times$), with the *largest single drop at base→SFT*: supervised fine-tuning on assistant transcripts installs most of the overconfidence, before any preference optimization (Fig. 1a). But the “collapses as context fills” framing is mostly a base-model ICL effect plus a floor: the within-context ratio *falls* across stages ($1.64 \rightarrow 0.47$), because aligned models start near a confidence floor and have nowhere left to drop. Within a fixed window, post-training is a downward *level shift* in baseline entropy, not a steeper collapse slope.

Finally, the mechanistic story does not survive a causal test. A boundary-detector SAE feature (Gemma Scope F90871, Lieberum et al., 2024) had been singled out as the regulator suppressed under fatigue. Clamping it back to its clean level on held-out fatigued cases moves entropy the *wrong* way ($0.442 \rightarrow 0.465$ vs. clean 0.293), leaves accuracy flat, and barely moves sycophancy; a random feature of similar magnitude perturbs entropy at least as much, and the feature’s decision-token suppression is sign-unstable across configurations. The single-feature mechanism is a context-dependent correlate, not a lever.

7 What context fatigue actually is

Across four model families and an organic-dialogue control, with attention dilution removed by construction: accumulation produces a real, monotone reallocation of attention and a real collapse of *confidence*, but no loss of accuracy, no loss of instruction adherence, and a within-context error signature that is the opposite of neglect. The confidence collapse is in-context comfort with a homogeneous task, installed by instruction tuning as a baseline level shift. “Context fatigue,” as the signatures measure it, is in-context learning, not decay. The genuine hazard is therefore not falling competence but *calibration*, and it is directly measurable (Fig. 2): on the DDXPlus streams logging per-token confidence ($n=154$ pooled cases, two models, both question orders), answer confidence rises $0.85 \rightarrow 0.95$ with fill ($r = +0.72 [+0.64, +0.79]$) at flat accuracy—and equally on *wrong* answers ($0.86 \rightarrow 0.95$; $r = +0.69 [+0.57, +0.78]$)—so the already-large confidently-wrong gap widens by ~ 10 points over a session, most consequential exactly where a homogeneous high-stakes stream makes the model surest.

Limitations. Per-condition n is bounded by the context window (tens of cases per session; we pool sessions); the OLMo sycophancy denominators are small and inconclusive; the instruction-canary null is a floor effect; and the WildChat figures are summary values from teacher-forced analyses. The accuracy claim is strongest for localized tasks—genuinely length-dependent single-task settings (where dilution is the point) are out of scope by design.

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